

Matrix Multiplication as the Sum of Outer Products

The transpose of a matrix times itself is equal to the sum of outer products created by the rows of the matrix. I prove this identity.

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In a [previous post](#), I used the following identity. Let $\mathbf{A}$ be an $N \times K$ matrix, and let $\mathbf{a}_n$ denote a K -dimensional row vector in $\mathbf{A}$. Then the following holds:

$$\mathbf{A}^\top \mathbf{A} = \sum_{n=1}^N \mathbf{a}_n^\top \mathbf{a}_n. \quad (1)$$

Note since $\mathbf{a}_n$ is a row vector, the operation $\mathbf{a}_n^\top \mathbf{a}_n$ is an outer product, not a dot product. Thus, the $K \times K$ matrix $\mathbf{A}^\top \mathbf{A}$ is the sum of N outer products. Let's prove this.

Let $\mathbf{0}$ denote a K -dimensional row vector of all zeros. We can write $\mathbf{A}$ as

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{0} \end{bmatrix} + \cdots + \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{a}_N \end{bmatrix} = \sum_{n=1}^N \begin{bmatrix} \mathbf{0} \\ \vdots \\ \mathbf{a}_n \\ \vdots \\ \mathbf{0} \end{bmatrix}. \quad (2)$$

This is nothing deep. We are writing $\mathbf{A}$ as the sum of N matrices, each of which has shape $N \times K$ and is all zeros except for the n -th row vector $\mathbf{a}_n$. Therefore we can write $\mathbf{A}^\top \mathbf{A}$ as

$$\mathbf{A}^\top \mathbf{A} = \mathbf{A}^\top \sum_{n=1}^N \begin{bmatrix} \mathbf{0} \\ \vdots \\ \mathbf{a}_n \\ \vdots \\ \mathbf{0} \end{bmatrix} = \sum_{n=1}^N \mathbf{A}^\top \begin{bmatrix} \mathbf{0} \\ \vdots \\ \mathbf{a}_n \\ \vdots \\ \mathbf{0} \end{bmatrix}, \quad (3)$$

because matrix multiplication distributes with respect to addition. It may not be obvious why this representation is helpful. Let's write it out explicitly:

$$\mathbf{A}^\top \mathbf{A} = \underbrace{\sum_{n=1}^N \left[\underbrace{\begin{bmatrix} a_{11} & \dots & a_{1n} & \dots & a_{1N} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{K1} & \dots & a_{Kn} & \dots & a_{KN} \end{bmatrix}}_{\mathbf{A}^\top} \underbrace{\begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nK} \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix}}_{\mathbf{B}^{(n)}} \right]}_{\mathbf{C}^{(n)}}. \quad (4)$$

In Eq. 4, each term inside the sum, labeled $\mathbf{C}^{(n)}$ is a matrix multiplication between a $K \times N$ matrix $\mathbf{A}^\top$ and an $N \times K$ matrix $\mathbf{B}^{(n)}$. This results in a $K \times K$ matrix $\mathbf{C}^{(n)}$. Using the definition of matrix multiplication, where $\mathbf{C}_{ij}^{(n)} = (\mathbf{A}_{i,:}^\top)^\top \mathbf{B}_{:,j}^{(n)}$, we see that each $\mathbf{C}^{(n)}$ is

$$\mathbf{C}^{(n)} = \begin{bmatrix} a_{1n}a_{n1} & \dots & a_{1n}a_{nK} \\ \vdots & \ddots & \vdots \\ a_{Kn}a_{n1} & \dots & a_{Kn}a_{nK} \end{bmatrix}. \quad (5)$$

And Eq. 5 is just the definition of the outer product between a column vector $\mathbf{a}_n^\top$ and a row vector $\mathbf{a}_n$,

$$\mathbf{D}^{(n)} = \mathbf{a}_n^\top \mathbf{a}_n. \quad (6)$$

Thus, we have proven Eq. 1. It's easy to extend this result to $\mathbf{A}\mathbf{A}^\top$. Just sum over K instead:

$$\mathbf{A}\mathbf{A}^\top = \sum_{k=1}^K \mathbf{a}_k \mathbf{a}_k^\top. \quad (7)$$

In this case, $\mathbf{a}_k$ is a column vector and the term $\mathbf{a}_k \mathbf{a}_k^\top$ is an outer product resulting in an $N \times N$ matrix.